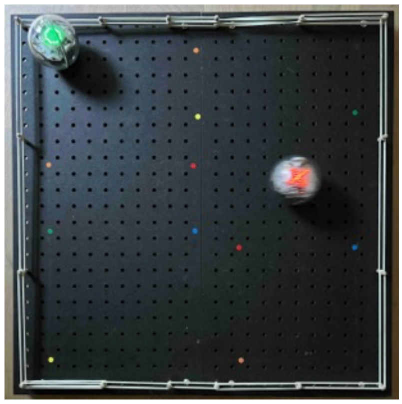
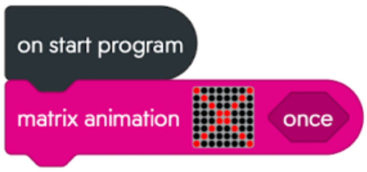
Run Robot Run

Robots give chase but may have to escape.



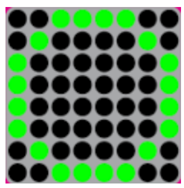
1) In Chrome, open https://edu.sphero.com/code
2) Connect your **Sphero BOLT** robot: click CONNECT
3) Select SPHERO BOLT then CONNECT
4) Your BOLT has a sticker with a code on it. Choose the matching BOLT from the list then Pair
5) Check your robot is showing the right image then CLOSE
6) Click **+ CREATE PROGRAM**. + CREATE PROGRAM
7) Input a name for your program. Leave **BLOCKS** selected. Select **SPHERO BOLT**, **deselect SPHERO BOLT+** then **CREATE**.
8) Add code for a **matrix animation** with *your own* design for a robot chaser in **red**. My example is a red **X**.



Click on the matrix grid to edit then + *to add a new 8x8 matrix, or* ✎ *to edit an existing one. Click* ✓ *to finish.*

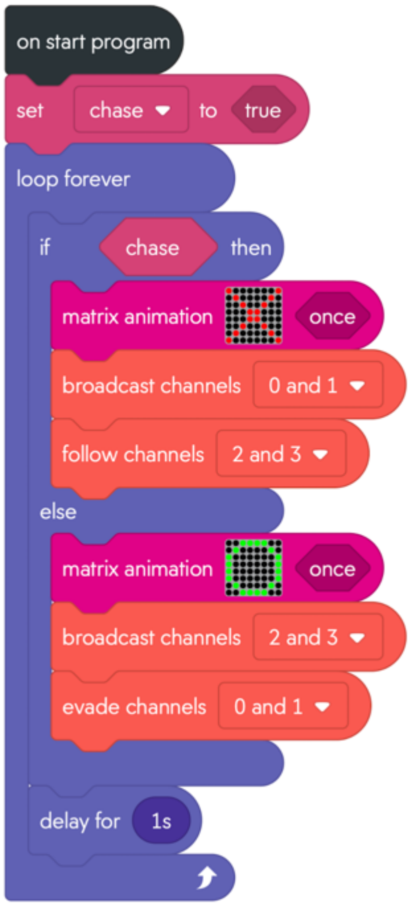
START *the code. Does it work?*

9) Click on the **matrix** grid and press + to add another 8x8 matrix in **green** for when your robot has to escape. My example is a green **O**.



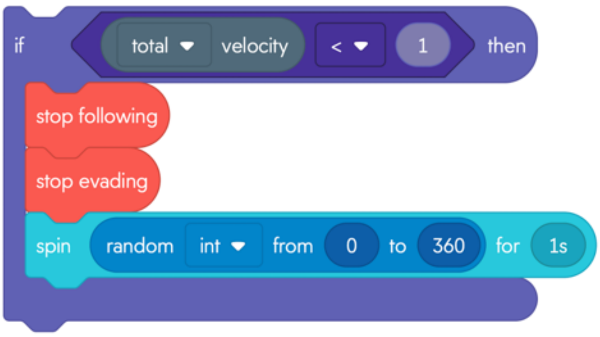
10) In **Variables**, create a new variable called **chase**. This is **boolean** because it will only be *true* or *false*.

Sphero robots communicate using infra-red light invisible to us. Chasers broadcast their position on channels 0 & 1, and those trying to escape on channels 2 & 3.



11) Add code (right) that **follows** or **evades** these channels depending on **if** it has to **chase** or escape. Click on *true/false* to toggle the value.

12) Sometimes the Sphero gets stuck on the sidelines. You can add code (left) after the delay, inside the loop, to randomly spin Sphero if it stops.



13) We can toggle between chase and escape modes **on collision** (but it won't detect glancing blows). The code (right) toggles between true and false.

*To **Save** your code go back to the main page. Click on the 3 dots (ellipsis) next to your program and download to a USB drive.*